

JKBOSE 2026

CLASS 11TH

STRUCTURE OF ATOM

LECTURE-10

CHEMISTRY

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Bhaichara
ON
TOP.



JKBose Wallah

Topics To Covered

1

2

ELECTRONIC CONFIGURATION OF ELEMENTS

3

STABILITY OF HALF-FILLED AND FULLY FILLED ORBITALS.

Important

ELECTRONIC CONFIGURATION



The distribution of electrons into orbitals of an atom is called its **electronic configuration**.

Representation of Electronic Configuration

a) nl^x notation

here, n = principal quantum number (shell).

l = Azimuthal quantum no. (subshell).

x = no. of e^- .

For example, Hydrogen ($Z=1$).



b) Orbital notation

It is more informative & each orbital is represented by a box.

For example, in case of Hydrogen.



Yeh Pata hona chahiye $\rightarrow$ Jigar.

1) ELECTRONIC CONFIGURATION^{of} ATOMS.

H (Z=1)	$1s^1$
He (Z=2)	$1s^2$
Li (Z=3)	$1s^2 2s^1$ or $[He] 2s^1$
Be (Z=4)	$1s^2 2s^2$ or $[He] 2s^2$
B (Z=5)	$1s^2 2s^2 2p^1$ or $[He] 2s^2 2p^1$
C (Z=6)	$1s^2 2s^2 2p^2$ or $[He] 2s^2 2p^2$
N (Z=7)	$1s^2 2s^2 2p^3$ or $[He] 2s^2 2p^3$
O (Z=8)	$1s^2 2s^2 2p^4$ or $[He] 2s^2 2p^4$

$\rightarrow$ order of energy.

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p \dots$$

$\rightarrow$ No. of electrons.

S-subshell $\rightarrow$ ↑↓ $\rightarrow 2e^-$

P-subshell $\rightarrow$ ↑↓ ↑↓ ↑↓ $\rightarrow 6e^-$
 $p_x \ p_y \ p_z$

d-subshell $\rightarrow$ ↑↓ ↑↓ ↑↓ ↑↓ ↑↓ $\rightarrow 10e^-$

$\rightarrow$ Noble gas.

- He
- Ne
- Ar

Z
2
10
18
...

F (Z=9)	$1s^2 2s^2 2p^5$ or [Ne] $2s^2 2p^5$
* Ne (Z=10)	<u>$1s^2 2s^2 2p^6$</u> or [Ne] $2s^2 2p^6$
Na (Z=11)	$1s^2 2s^2 2p^6 3s^1$ or [Ne] $3s^1$
Mg (Z=12)	<u>$1s^2 2s^2 2p^6 3s^2$</u> or [Ne] $3s^2$
Al (Z=13)	$1s^2 2s^2 \underline{2p^6} 3s^2 3p^1$ or [Ne] $3s^2 3p^1$
Si (Z=14)	
P (Z=15)	
S (Z=16)	
Cl (Z=17)	
* Ar (Z=18)	

K

19

20

21

Sc (Z=21)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$ OR $[Ar].4s^2 3d^1$
Ti (Z=22)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$ OR $[Ar] 4s^2 3d^2$
V (Z=23)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$ OR $[Ar] 4s^2 3d^3$
★ Cr (Z=24)	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4 \rightarrow X 4s^1 3d^5$ or $[Ar] 4s^1 3d^5$
Mn (Z=25)	or $[Ar] 4s^2 3d^5$
Fe (Z=26)	or $[Ar] 4s^1 3d^6$
Co (Z=27)	or $[Ar] 4s^2 3d^7$
Ni (Z=28)	or $[Ar] 4s^2 3d^8$
★ Cu (Z=29)	or $[Ar] 4s^2 3d^9 \rightarrow X 4s^1 3d^{10}$
Zn (Z=30)	or $[Ar] 4s^2 3d^{10}$

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p \dots$

ELECTRONIC CONFIGURATION

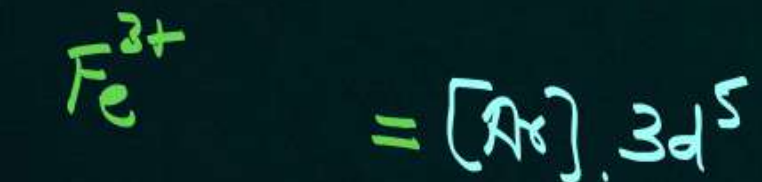
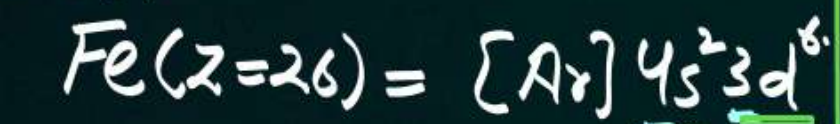


KAM KI BAAT.

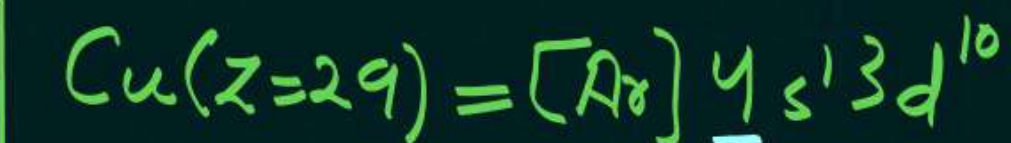
2) ELECTRONIC CONFIGURATION OF IONS $\rightarrow$



Sol: $\rightarrow$



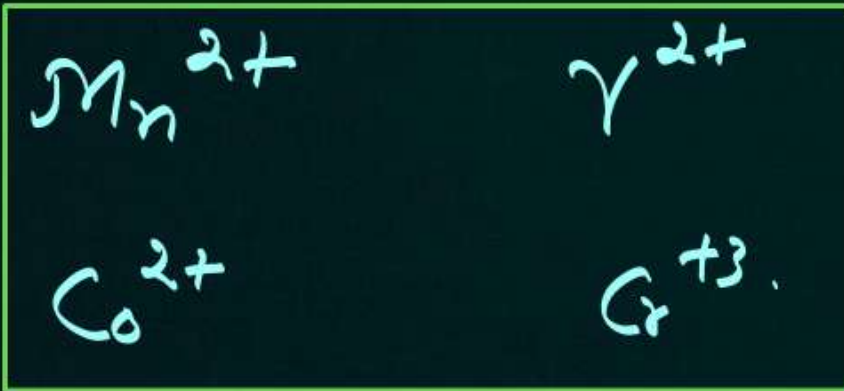
Sol: $\rightarrow$



Jab neutral atom se e^- nikalte hain, to hum us shell se nikalte hai jiska principal quantum no. sabse zyada hota. or $\{n \text{ value}\}$
or outermost shell

HOMEWORK FOR COMMUNITY.

Q. Write the. electronic config.
for.



	ELECTRON CONFIGURATION OF ELEMENT
1s ¹	
1s ²	
1s ² 2s ¹	
1s ² 2s ²	
1s ² 2s ² 2p ¹	
1s ² 2s ² 2p ²	
1s ² 2s ² 2p ³	
1s ² 2s ² 2p ⁴	
1s ² 2s ² 2p ⁵	
1s ² 2s ² 2p ⁶	
1s ² 2s ² 2p ⁶ 3s ¹	
1s ² 2s ² 2p ⁶ 3s ²	
1s ² 2s ² 2p ⁶ 3s ² 3p ¹	
1s ² 2s ² 2p ⁶ 3s ² 3p ²	
1s ² 2s ² 2p ⁶ 3s ² 3p ³	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁴	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁵	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ¹	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ²	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ¹	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ²	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ³	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ¹ 3d ⁵	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁵	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁶	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁷	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ⁸	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ¹ 3d ¹⁰	
1s ² 2s ² 2p ⁶ 3s ² 3p ⁶ 4s ² 3d ¹⁰	

QUESTION



How many electrons can be present in a subshell with $l = 2$?

1 10

2 6

3 8

4 2

$$l = 2$$

$$l = 0, 1, 2$$

s

p

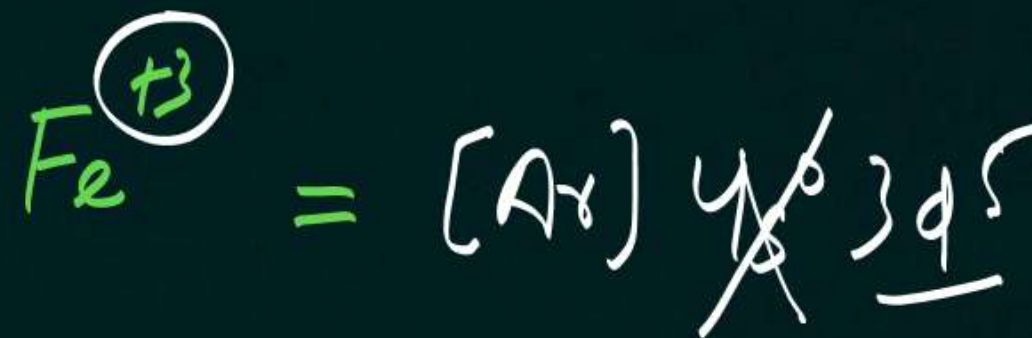
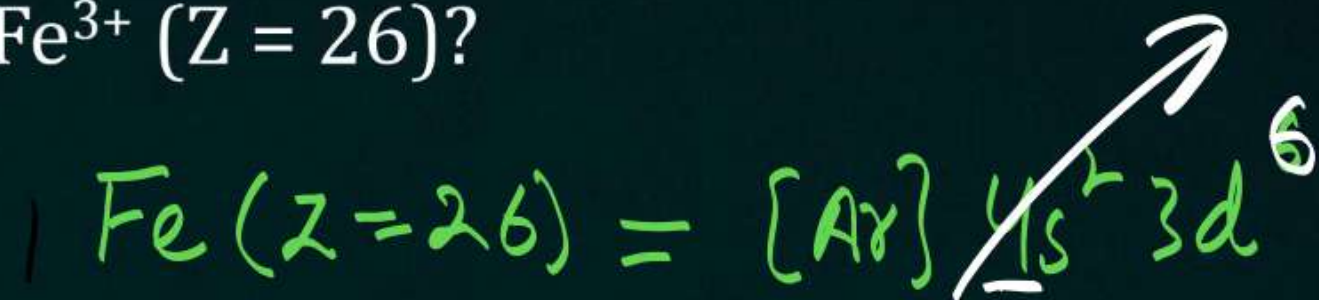
d



QUESTION



What is the electronic configuration of Fe^{3+} ($Z = 26$)?



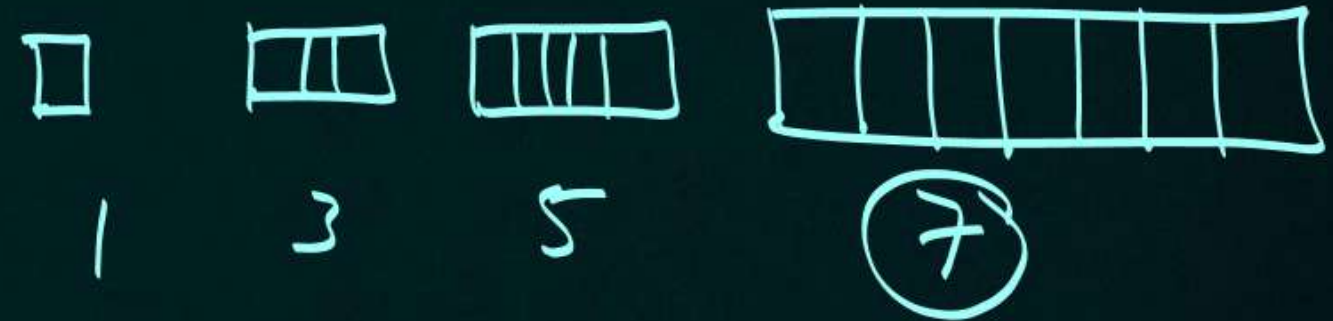
QUESTION



The maximum number of electrons in a subshell with $l = 3$ is:

$$l = 0, 1, 2, 3$$

s p d **f**



1

6

2

10

3

14

4

18

- 1) Wahid Rasool.
- 2) UZAIR SUNAIL
- ③ Aafreena Fayaz.

REVISE KAR KE Aana.

Thank
You